

Tapped Indoor Planter

Group 102-3

Tapped Indoor Planter

Need: Taking care of indoor plants can be a pain

- Current planter drainage trays cause root rot, attract pests, and can spill over and cause water damage.
- Drainage trays must be regularly emptied, which is inefficient.

Market Size:

- Demographic: Millennials and Gen Z (ages 18-44) -> working age
- ~66% of U.S. households have at least one indoor plant (~89 million households)
- Indoor plant market size: [\\$13.61B](#) (projected 20% growth over 5 years)

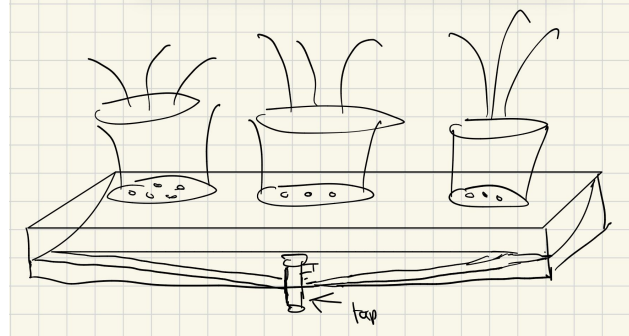
Societal Impact: Indoor plants improve indoor air quality and mental health

- Make it easier to have houseplants
- More accessible to people with mobility restrictions (don't have to pick up pots to empty drainage trays)

Design Concept

Working Principle: dual-chamber system, top has roots and dirt, bottom holds the water runoff in a reservoir which has the mechanical valve, allowing the user to control the out-flow. At the interface between the two is a perforated separation plate.

Current Concept: See image ->



Tentative Materials & Manufacturing: PETG FDM 3D printing for prototyping, fun colors are possible which can transfer over into an injection molded design. Metal can be used as accent and for spigot design, CNC milled stainless for the most part. (Clay?)

Questions

- O-Rings? How to work with water with good seals?
- Modularity? Had the idea of stacking the flowpath of many different planters so it could look like a cool (filtered?) waterfall from planter to planter
- How should we design the tap part of this for it not only to be easy to use but also possibly automatized (have electronics) in a “future revision?”

Summary of Design Changes:

- Added an intermediate plate to help reduce the sound of water droplets
- Moved the spigot hole to the side to allow for easier emptying and potential planter tiering
- Changed the slope of bottom to 2° front/back and 4° side-middle to maximize internal volume while maintaining proper water outflow
- Added O-ring groove on outer edge
- Added ingroove into top plate for better planter seeding, with less area coverage on bottom of planter, blocking water

Topics of Needed Help:

- O-Ring?
- Conformal Coating?
- Spigot options and pre-threading into 3d print?
- Press-fit top plate?
- Nylon Tape?
- How to make sure there is as little pooling between the pot and plate?
- Print the droplet mitigator separate?

notes

Smaller standoffs on top plate to prevent water accumulation

Potential tube attached to spigot to drain water to a specific location